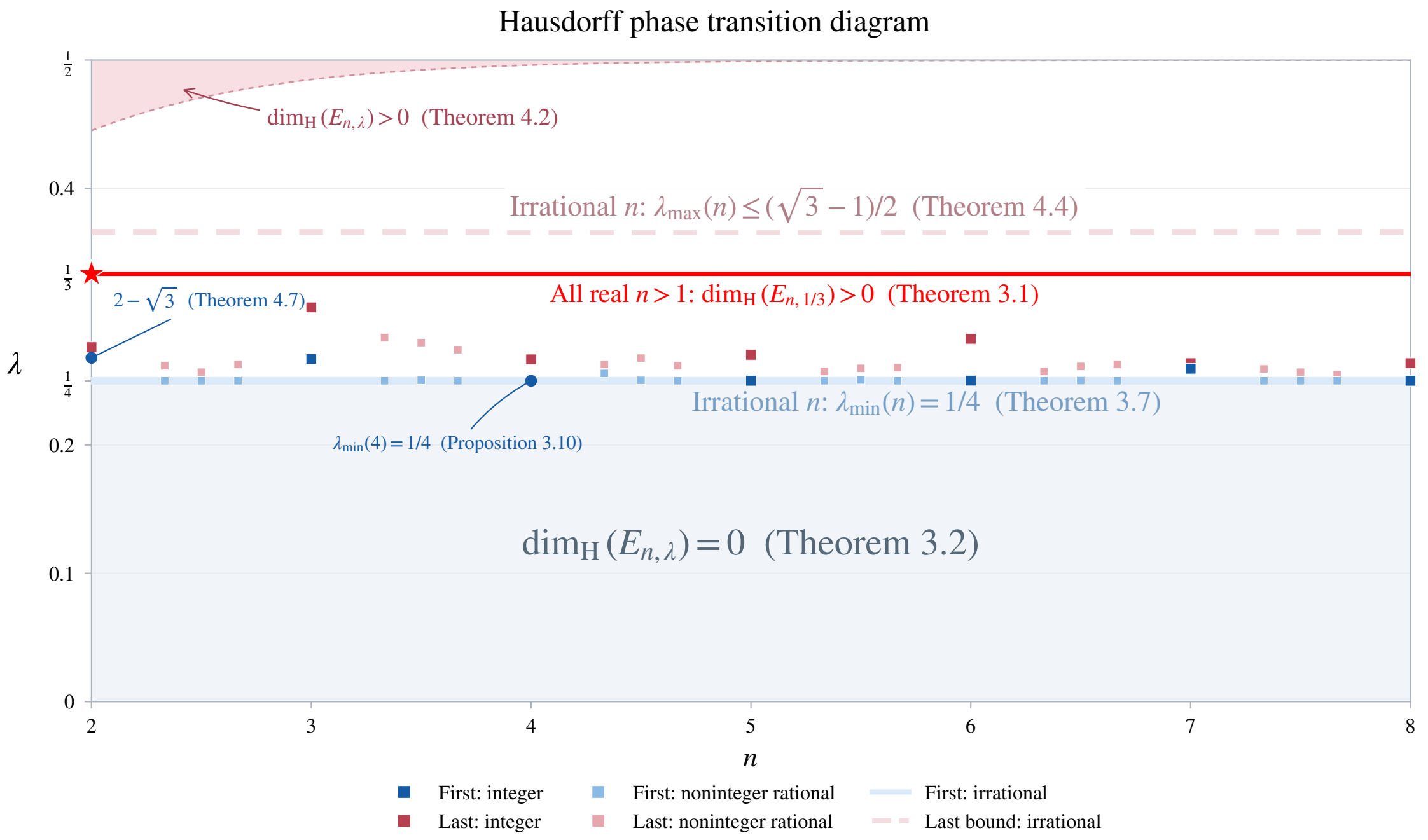
Hausdorff phase transition diagram
$\frac{1}{2}$
0.4
$\frac{1}{3}$
$\frac{1}{4}$
0.2
0.1
0
λ
$\dim_H(E_{n,\lambda}) > 0$ (Theorem 4.2)
Irrational n: $\lambda_{\max}(n) \le (\sqrt{3}-1)/2$ (Theorem 4.4)
All real $n > 1$: $\dim_H(E_{n,1/3}) > 0$ (Theorem 3.1)
$2-\sqrt{3}$ (Theorem 4.7)
Irrational n: $\lambda_{\min}(n) = 1/4$ (Theorem 3.7)
$\lambda_{\min}(4) = 1/4$ (Proposition 3.10)
$\dim_H(E_{n,\lambda}) = 0$ (Theorem 3.2)
2
3
4
5
6
7
8
n
First: integer
Last: integer
First: noninteger rational
Last: noninteger rational
First: irrational
Last bound: irrational
Circles: exact values | Squares: numerical estimates, not rigorous bounds